Babu Banarasi Das University

Case Study
ON

Netflix Data Anlysis using Hadoop

Submitted To:-
Mr. Vikash (IBM)

Submitted By:-
Name – Akhil Raj Patel
Section – BCADS21
Roll Number – 1240258054

PROJECT

Netflix Dataset MapReduce Practical

- **Architecture:**

  - **The architecture of this project follows a standard Big Data pipeline using Hadoop ecosystem components.**

- **Data Flow:**

  - **Raw Dataset → HDFS → MapReduce Processing → Processed Data → Hive Analysis → Insights**

- **Explanation:**

  - **Raw dataset (netflix data) is first stored in HDFS**
  - **MapReduce is used to process the data and perform analysis such as counting Movies and TV Shows**
  - **Processed data is stored back in HDFS**
  - **Hive is used to perform analytical queries**
  - **Final insights are generated from the results1**

- **Tools Used:**

  - **Hadoop Distributed File System (HDFS)**
  - **MapReduce (Hadoop Streaming using Python)**
  - **Hive (for data analysis)**
  - **Linux Terminal**
  - **ChatGPT (Generative AI tool)**

- **Technologies:**

  - **Python (for mapper and reducer)**
  - **CSV dataset**
  - **Hadoop Ecosystem**

➢ Use of Generative AI in the Project:

- **In this project, Generative AI tools were used to assist in development and understanding of Big Data concepts.**

➢ Areas Where AI Was Used:

1. **Code Generation**
   - **Generative AI was used to generate MapReduce programs (mapper and reducer).**
2. **Debugging**
   - **Errors in code were identified and fixed using AI assistance.**
3. **Query Writing**
   - **Hive queries were generated and optimized using AI.**
4. **Concept Understanding**
   - **AI helped understand HDFS, MapReduce, and Hive**

➢ Example Prompt Used:

- **Write a Python MapReduce program to count number of Movies and TV Shows.**

➢ Benefits of Using AI:

- **Faster development**
- **Reduced coding errors**
- **Better understanding of concepts**
- **Improved efficiency**

➢ Validation:

- **All AI-generated code was manually tested and verified before execution.**

➢ Dataset Description:

- **The dataset used in this project is Netflix dataset containing information about movies and TV shows available on Netflix.**

➢ Dataset Structure:

- **The dataset consists of the following columns:**
  - **show_id – Unique ID**
  - **type – Movie or TV Show**
  - **title – Name of content**
  - **director – Director name**
  - **cast – Actors**
  - **country – Country of production**
  - **date_added – Date added to Netflix**
  - **release_year – Release year**
  - **rating – Age rating**
  - **duration – Duration**
  - **listed_in – Genre**

➢ Dataset Source:

- **The dataset is a combination of real-world movie data and custom entries created for project demonstration.**

➢ Purpose of Dataset:

- **Count Movies vs TV Shows**
- **Analyze content distribution**
- **Identify trends**
- **Perform data processing using MapReduce**
- **Generate insights using Hive**

➢ Sample Data:

```
show_id,type,title,director,cast,country,date_added,release_year,rating,duration,listed_in,description
s1,Movie,Dick Johnson Is Dead,Kirsten Johnson,,United States,"September 25, 2021",2020,PG-13,90 min,Documentaries,"As her fa$
s2,TV Show,Blood & Water,,"Ama Qamata, Khosi Ngema, Gail Mabalane, Thabang Molaba, Dillon Windvogel, Natasha Thahane, Arno G$
s3,TV Show,Ganglands,Julien Leclercq,"Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabiha Akkari, Sofia Lesaffre, Salim Kechiou$
s4,TV Show,Jailbirds New Orleans,,,,"September 24, 2021",2021,TV-MA,1 Season,"Docuseries, Reality TV","Feuds, flirtations an$
s5,TV Show,Kota Factory,,"Mayur More, Jitendra Kumar, Ranjan Raj, Alam Khan, Ahsaas Channa, Revathi Pillai, Urvi Singh, Arun$
s6,TV Show,Midnight Mass,Mike Flanagan,"Kate Siegel, Zach Gilford, Hamish Linklater, Henry Thomas, Kristin Lehman, Samantha $
s7,Movie,My Little Pony: A New Generation,"Robert Cullen, José Luis Ucha","Vanessa Hudgens, Kimiko Glenn, James Marsden, Sof$
s8,Movie,Sankofa,Haile Gerima,"Kofi Ghanaba, Oyafunmike Ogunlano, Alexandra Duah, Nick Medley, Mutabaruka, Afemo Omilami, Re$
s9,TV Show,The Great British Baking Show,Andy Devonshire,"Mel Giedroyc, Sue Perkins, Mary Berry, Paul Hollywood",United King$
s10,Movie,The Starling,Theodore Melfi,"Melissa McCarthy, Chris O'Dowd, Kevin Kline, Timothy Olyphant, Daveed Diggs, Skyler G$
s11,TV Show,"Vendetta: Truth, Lies and The Mafia",,,,"September 24, 2021",2021,TV-MA,1 Season,"Crime TV Shows, Docuseries, I$
s12,TV Show,Bangkok Breaking,Kongkiat Komesiri,"Sukollawat Kanarot, Sushar Manaying, Pavarit Mongkolpisit, Sahajak Boonthana$
s13,Movie,Je Suis Karl,Christian Schwochow,"Luna Wedler, Jannis Niewöhner, Milan Peschel, Edin Hasanović, Anna Fialová, Marl$
s14,Movie,Confessions of an Invisible Girl,Bruno Garotti,"Klara Castanho, Lucca Picon, Júlia Gomes, Marcus Bessa, Kiria Malh$
s15,TV Show,Crime Stories: India Detectives,,,,"September 22, 2021",2021,TV-MA,1 Season,"British TV Shows, Crime TV Shows, D$
```

➢ Conclusion:

- **In this project, a complete Big Data pipeline was implemented using Hadoop ecosystem.**
- **The Netflix dataset was stored in HDFS, processed using MapReduce to count Movies and TV Shows, and analyzed using Hive to generate insights.**
- **This project helped in understanding distributed data processing and Big Data analytics.**
- **Generative AI helped in code generation, debugging, and understanding concepts.**

❖ **Generative AI was used not only for code generation but also as a learning assistant to understand Big Data workflows and improve overall efficiency.**

➢ <u>Question-1:-</u> Movie Rating Analysis using Hadoop Streaming.

❖ **Aim: - "To analyze Netflix dataset and count Movies and TV Shows using MapReduce ."**

Step-01: - Create **a "CSV" file.**

- **Cat > netflix_titles.csv**

```
[cloudera@quickstart ~]$ cat netflix_titles.csv

show_id,type,title,director,cast,country,date_added,release_year,rating,duration,listed_in,description
s1,Movie,Dangal,Nitesh Tiwari,"Aamir Khan, Sakshi Tanwar, Fatima Sana Shaikh",India,December 31, 2016,2016,TV-PG,161 min,"Dramas, International
Movies,Sports Movies","Former wrestler Mahavir Singh Phogat trains his daughters to win gold at the Commonwalth Games."
s2,Movie,3 Idiots,Rajkumar Hirani,"Aamir Khan, Madhavan, Mona Singh",India,December 31, 2016,2009,PG-13,171 min,"Comedies, Dramas, International
 Movies","Two friends are searching for their long lost companion. They revisit their college days and recall the memories."
s3,Movie,Baahubali: The Beginning,S.S. Rajamouli,"Prabhas, Rana Daggubati, Anushka Shetty",India,December 31, 2016,2015,TV-14,159 min,"Action,
& Adventure, Dramas, International Movies","In ancient India, an adventurous and daring man becomes involved in a decades-old feud between two warring peoples."
s4,Movie,Baahubali 2: The Conclusion,S.S. Rajamouli,"Prabhas, Rana Daggubati, Anushka Shetty",India,December 31, 2016,2017,TV-14,167 min,"Action
& Adventure, Dramas, International Movies","When Shiva, the son of Bahubali, learns about his heritage, he begins to look for answers."
s5,Movie,PK,Rajkumar Hirani,"Aamir Khan, Anushka Sharma, Sanjay Dutt",India,December 31, 2016,2014,PG,153 min,"Comedies, Dramas, Internati
ional Movies","An alien on Earth loses the only device he can use to communicate with his spaceship."
s6,Movie,Chhichhore,Nitesh Tiwari,"Sushant Singh Rajput, Shraddha Kapoor, Varun Sharma",India,August 1, 2020,2019,TV-14,143 min,"Comedies,
Dramas, International Movies","A group of engineering students look to an unusual adult for help."
s7,Movie,Sultan,Ali Abbas Zafar,"Salman Khan, Anushka Sharma, Randeep Hooda",India,December 31, 2016,2016,TV-14,170 min,"Dramas, Inter-
ational Movies,Sports Movies","Sultan Ali Khan is a wrestling champion who decides to fight his way back to the top."
s8,Movie,Pad Man,R. Balki,"Akshay Kumar, Sonam Kapoor, Radhika Apte",India,December 31, 2016,2018,TV-14,140 min,"Comedies, Dramas, Inter
ational Movies","A man invents a low-cost sanitary napkin machine to benefit the women in his village."
s9,Movie,Sanju,Rajkumar Hirani,"Ranbir Kapoor, Paresh Rawal, Anushka Sharma",India,July 5, 2018,2018,TV-14,161 min,"Biographical Movies,
Dramas, International Movies","The life of actor Sanjay Dutt, who has had a tumultuous personal life."
s10,Movie,War,Siddharth Anand,"Hrithik Roshan, Tiger Shroff, Vaani Kapoor",India,March 1, 2020,2019,TV-14,151 min,"Action & Adventure,
International Movies,Thrillers","An Indian soldier chases after his mentor who has gone rogue following an unexpected kill."
s11,Movie,Uri: The Surgical Strike,Aditya Dhar,"Vicky Kaushal, Paresh Rawal, Mohit Raina",India,May 11, 2019,2019,TV-14,138 min,"Ac-
on &Adventure, Dramas, International Movies","Indian army special forces execute a covert operation."
s12,Movie,Kesari,Anurag Singh,"Akshay Kumar, Parineeti Chopra, Suvinder Vicky",India,March 1, 2020,2019,TV-14,153 min,"Action & Adventure,
Dramas, International Movies","A story of Havildar Ishar Singh and his 21 men who fought bravely."
s13,Movie,Queen,Vikas Bahl,"Kangana Ranaut, Rajkummar Rao, Lisa Haydon",India,July 15, 2018,2013,PG-13,146 min,"Comedies, Dramas,
national Movies","A girl from Delhi goes on her honeymoon alone after her fiance calls off the wedding."
s14,Movie,Zindagi Na Milegi Dobara,Zoya Akhtar,"Hrithik Roshan, Farhan Akhtar, Abhay Deol",India,December 31, 2016,2011,PG-13,155 mi-
"Comedies, Dramas, International Movies","Three friends go on a road trip that changes their lives forever."
s15,Movie,Tamasha,Imtiaz Ali,"Ranbir Kapoor, Deepika Padukone, Shahid Kapoor",India,February 1, 2021,2015,PG-13,139 min,"Comedies,
Dramas, International Movies","A man unhappy with his life meets a wild young woman."
s16,Movie,Stree,Amar Kaushik,"Rajkummar Rao, Shraddha Kapoor, Pankaj Tripathi",India,November 2, 2018,2018,TV-14,128 min,"Comedies, Horro-
Movies, International Movies","In the small town of Chanderi, men live in fear of Stree."
s17,Movie,Kabir Singh,Sandeep Reddy Vanga,"Shahid Kapoor, Kiara Advani",India,July 15, 2020,2019,TV-MA,173 min,"Dramas, International
Movies,Romantic Movies","An intense portrayal of love, anger and self-destruction."
s18,Movie,Gully Boy,Zoya Akhtar,"Ranveer Singh, Alia Bhatt, Siddhant Chaturvedi",India,April 14, 2019,2019,TV-14,154 min,"Dramas,
Music & Musicals, International Movies","A coming-of-age story of a street rapper from Mumbai."
s19,Movie,Ludo,Anurag Basu,"Abhishek Bachchan, Rajkummar Rao, Fatima Sana Shaikh",India,April 12, 2021,2020,TV-MA,150 min,"Comedies,
Crime Movies, Dramas","A dark comedy that interweaves the stories of four people."
s20,Movie,Bhool Bhulaiyaa,Anees Bazmee,"Akshay Kumar, Vidya Balan, Shiney Ahuja",India,December 31, 2016,2007,PG-13,159 min,"Comedies, Horr-
Movies, International Movies","A case of mistaken identity leads to hilarious situations."

[cloudera@quickstart ~]$
```

Step-02:- Upload Dataset

- **hdfs dfs -mkdir /user/cloudera/akhil**
- **hdfs dfs -mkdir /user/cloudera/akhil/netflix**

```
[cloudera@quickstart akhil]$ hdfs dfs -mkdir /user/cloudera/akhil
[cloudera@quickstart akhil]$ hdfs dfs -mkdir /user/cloudera/akhil/netflix
[cloudera@quickstart akhil]$ hdfs dfs -put /home/cloudera/akhil/netflix_titles.c
sv /user/cloudera/akhil/netflix/
[cloudera@quickstart akhil]$ hdfs dfs -ls /user/cloudera/akhil/netflix
Found 1 items
-rw-r--r--   1 cloudera supergroup    3399671 2026-04-22 23:44 /user/cloudera/ak
hil/netflix/netflix_titles.csv
[cloudera@quickstart akhil]$
```

Step-03:- Load the "CSV" on Cloudera's Server

- **hdfs dfs -put /home/cloudera/akhil/netflix_titles.csv /user/cloudera/akhil/netflix/**
- **hdfs dfs -ls /user/cloudera/akhil/netflix**

```
[cloudera@quickstart akhil]$ hdfs dfs -put /home/cloudera/akhil/netflix_titles.c
sv /user/cloudera/akhil/netflix/
[cloudera@quickstart akhil]$ hdfs dfs -ls /user/cloudera/akhil/netflix
Found 1 items
-rw-r--r--   1 cloudera supergroup    3399671 2026-04-22 23:44 /user/cloudera/ak
hil/netflix/netflix_titles.csv
```

Step-04:- Create a mapper file using "nano"

- **nano mapper.py**

```
#!/usr/bin/env python
import sys

for line in sys.stdin:
    if "show_id" in line:
        continue
    data = line.strip().split(',')
    type_data = data[1]
    print("%s\t1" % type_data)
```

```
#!/usr/bin/env python
import sys

for line in sys.stdin:
    if "show_id" in line:
        continue
    data = line.strip().split(',')
    type_data = data[1]
    print("%s\t1" % type_data)
```

Step-05:- Create a reducer file using "nano"

- **nano reducer.py**

Code :-

```
#!/usr/bin/env python
import sys
current = None
count = 0

for line in sys.stdin:
key, value = line.strip().split('\t')
value = int(value)

if current == key:
count += value
else:
If current:
print("%s\t%d" % (current, count))
current = key
count = value

if current:
print("%s\t%d" % (current, count))
```

```
import sys

current = None
count = 0

for line in sys.stdin:
    key, value = line.strip().split('\t')
    value = int(value)

    if current == key:
        count += value
    else:
        if current:
            print("%s\t%d" % (current, count))
        current = key
        count = value

if current:
    print("%s\t%d" % (current, count))
```

Step-06:- Make executable

- **chmod +x mapper.py reducer.py**

```
[cloudera@quickstart akhil]$ chmod +x mapper.py reducer.py
[cloudera@quickstart akhil]$ ls
akhil.txt  mapper.py  netflix_titles.csv  reducer.py
[cloudera@quickstart akhil]$ ls -l
total 3332
-rw-rw-r-- 1 cloudera cloudera      32 Mar 28 06:14 akhil.txt
-rwxrwxr-x 1 cloudera cloudera     190 Apr 22 23:52 mapper.py
-rwxrwxrwx 1 cloudera cloudera 3399671 Apr 22 23:30 netflix_titles.csv
-rwxrwxr-x 1 cloudera cloudera     368 Apr 22 23:59 reducer.py
[cloudera@quickstart akhil]$
```

Step-07:- Copy "netflix_titles.csv" file, from /home/cloudera to /home/cloudera/mapreduce

- **cp /home/cloudera/netflix_titles.csv /home/cloudera/mapreduce/netflix_titles.csv**

```
[cloudera@quickstart mapreduce]$ cp /home/cloudera/movie.csv /home/cloudera/mapreduce/movie.csv
[cloudera@quickstart mapreduce]$ ls
mapper.py  movie.csv  reducer.py
[cloudera@quickstart mapreduce]$
```

Step-08:- Local test

- **cat netflix_titles.csv | python mapper.py**
- **cat netflix_titles.csv | python mapper.py | sort | python reducer.py**

```
Movie   1
TV Show 1
Movie   1
Movie   1
Movie   1
Movie   1
TV Show 1
Movie   1
Movie   1
TV Show 1
Movie   1
TV Show 1
Movie   1
Movie   1
Movie   1
Movie   1
Movie   1
Movie   1
Movie   1
Movie   1
Movie   1
TV Show 1
TV Show 1
TV Show 1
Movie   1
Movie   1
TV Show 1
Movie   1
Movie   1
TV Show 1
Movie   1
Movie   1
Movie   1
[cloudera@quickstart akhil]$ cat netflix_titles.csv | python mapper.py | sort |
python reducer.py
Movie   6131
TV Show 2676
William Wyler   1
[cloudera@quickstart akhil]$
```

Step-09:- Run Hadoop Job

```
hadoop jar /usr/lib/hadoop-mapreduce/hadoop-streaming.jar \
-files mapper.py,reducer.py \
-input /user/cloudera/akhil/netflix/netflix_titles.csv \
-output /user/cloudera/akhil/netflix/output \
-mapper "python mapper.py" \
-reducer "python reducer.py"
```

```
[cloudera@quickstart akhil]$ hadoop jar /usr/lib/hadoop-mapreduce/hadoop-streami
ng.jar \
> -files mapper.py,reducer.py \
> -input /user/cloudera/akhil/netflix/netflix_titles.csv \
> -output /user/cloudera/akhil/netflix/output \
> -mapper "python mapper.py" \
> -reducer "python reducer.py"
packageJobJar: [] [/usr/jars/hadoop-streaming-2.6.0-cdh5.4.2.jar] /tmp/streamjob
7073345981166337256.jar tmpDir=null
26/04/23 00:12:04 INFO client.RMProxy: Connecting to ResourceManager at /0.0.0.0
:8032
26/04/23 00:12:04 INFO client.RMProxy: Connecting to ResourceManager at /0.0.0.0
:8032
26/04/23 00:12:05 INFO mapred.FileInputFormat: Total input paths to process : 1
26/04/23 00:12:05 INFO mapreduce.JobSubmitter: number of splits:2
26/04/23 00:12:06 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_17
76925353456_0001
26/04/23 00:12:06 INFO impl.YarnClientImpl: Submitted application application_17
76925353456_0001
26/04/23 00:12:06 INFO mapreduce.Job: The url to track the job: http://quickstar
t.cloudera:8088/proxy/application_1776925353456_0001/
26/04/23 00:12:06 INFO mapreduce.Job: Running job: job_1776925353456_0001
26/04/23 00:12:16 INFO mapreduce.Job: Job job_1776925353456_0001 running in uber
 mode : false
26/04/23 00:12:16 INFO mapreduce.Job:  map 0% reduce 0%
26/04/23 00:12:23 INFO mapreduce.Job:  map 50% reduce 0%
26/04/23 00:12:24 INFO mapreduce.Job:  map 100% reduce 0%
26/04/23 00:12:29 INFO mapreduce.Job:  map 100% reduce 100%
26/04/23 00:12:29 INFO mapreduce.Job: Job job_1776925353456_0001 completed succe
ssfully
26/04/23 00:12:29 INFO mapreduce.Job: Counters: 49
        File System Counters
                FILE: Number of bytes read=93446
                FILE: Number of bytes written=528444
                FILE: Number of read operations=0
                FILE: Number of large read operations=0
                FILE: Number of write operations=0
                HDFS: Number of bytes read=3404032
```

Step-10:- View Output ;-

```
hdfs dfs -cat /user/cloudera/akhil/movie/output/part-00000
```

```
[cloudera@quickstart akhil]$ hdfs dfs -cat /user/cloudera/akhil/netflix/output/part-00000
Movie   6131
TV Show 2676
William Wyler   1
[cloudera@quickstart akhil]$
```

- **<u>Question-2:-</u> To perform data analysis on processed netflix dataset using Hive.**

  - **Aim: - "To analyze Netflix dataset using Hive queries."**

Step-01:- Open Hive

- **hive**

```
[cloudera@quickstart akhil]$ hive

Logging initialized using configuration in file:/etc/hive/conf.dist/hive-log4j.p
roperties
WARNING: Hive CLI is deprecated and migration to Beeline is recommended.
hive>
```

Step-02:- Create table on hive

```
CREATE TABLE netflix (
show_id STRING,
type STRING,
title STRING,
director STRING,
cast_name STRING,
country STRING,
date_added STRING,
release_year INT,
rating STRING,
duration STRING,
listed_in STRING)
ROW FORMAT DELIMITED
FIELDS TERMINATED BY ','
STORED AS TEXTFILE;
```

```
hive> CREATE TABLE netflix (
    > show_id STRING,
    > type STRING,
    > title STRING,
    > director STRING,
    > cast_name STRING,
    > country STRING,
    > date_added STRING,
    > release_year INT,
    > rating STRING,
    > duration STRING,
    > listed_in STRING
    > )
    > ROW FORMAT DELIMITED
    > FIELDS TERMINATED BY ','
    > STORED AS TEXTFILE;
OK
Time taken: 1.724 seconds
hive>
```

Step-03:- Load Data from HDFS

- LOAD DATA INPATH '/user/cloudera/akhil/netflix_titles.csv' INTO TABLE netflix;

```
hive> LOAD DATA INPATH '/user/cloudera/akhil/netflix_titles.csv'
    > INTO TABLE netflix;
Loading data to table default.netflix
Table default.netflix stats: [numFiles=1, totalSize=3399671]
OK
Time taken: 1.007 seconds
hive>
```

Step-04:- View Data

- SELECT * FROM netflix LIMIT 10;

```
hive> SELECT * FROM netflix LIMIT 10;
OK
show_id type    title   director        cast    country date_added      NULL    rating  duration        listed_in
s1      Movie   Dick Johnson Is Dead    Kirsten Johnson         United States   "September 25   NULL    2020    PG-13   90 mi
n
s2      TV Show Blood & Water           "Ama Qamata      Khosi Ngema     Gail Mabalane  NULL     Dillon Windvogel        Nata
sha Thahane      Arno Greeff
s3      TV Show Ganglands       Julien Leclercq "Sami Bouajila   Tracy Gotoas    Samuel Jouy    NULL     Sofia Lesaffre  Sali
m Kechiouche     Noureddine Farihi
s4      TV Show Jailbirds New Orleans                           "September 24   NULL    2021    TV-MA   1 Season
s5      TV Show Kota Factory            "Mayur More      Jitendra Kumar  Ranjan Raj     NULL     Ahsaas Channa   Revathi Pill
ai       Urvi Singh
s6      TV Show Midnight Mass   Mike Flanagan   "Kate Siegel     Zach Gilford    Hamish Linklater       NULL     Kristin Lehm
an       Samantha Sloyan         Igby Rigney
s7      Movie   My Little Pony: A New Generation        "Robert Cullen   José Luis Ucha"        "Vanessa Hudgens         Kimi
ko Glenn        NULL     Sofia Carson    Liza Koshy      Ken Jeong
s8      Movie   Sankofa Haile Gerima    "Kofi Ghanaba    Oyafunmike Ogunlano     Alexandra Duah NULL     Mutabaruka      Afem
o Omilami        Reggie Carter
s9      TV Show The Great British Baking Show   Andy Devonshire "Mel Giedroyc    Sue Perkins     Mary Berry     NULL    Unite
d Kingdom       "September 24    2021"
Time taken: 0.664 seconds, Fetched: 10 row(s)
hive>
```

Step-05:- Count Movies vs TV Shows

- SELECT type, COUNT(*) FROM netflix
  GROUP BY type;

```
hive> SELECT type, COUNT(*)
    > FROM netflix
    > GROUP BY type;
Query ID = cloudera_20260423004444_f3f096aa-f901-49ba-b4ef-2f5eb79a69fc
Total jobs = 1
Launching Job 1 out of 1
Number of reduce tasks not specified. Estimated from input data size: 1
In order to change the average load for a reducer (in bytes):
  set hive.exec.reducers.bytes.per.reducer=<number>
In order to limit the maximum number of reducers:
  set hive.exec.reducers.max=<number>
In order to set a constant number of reducers:
  set mapreduce.job.reduces=<number>
Starting Job = job_1776925353456_0002, Tracking URL = http://quickstart.cloudera:8088/proxy/application_1776925353456_0002/
Kill Command = /usr/lib/hadoop/bin/hadoop job  -kill job_1776925353456_0002
Hadoop job information for Stage-1: number of mappers: 1; number of reducers: 1
2026-04-23 00:44:35,445 Stage-1 map = 0%,  reduce = 0%
2026-04-23 00:44:41,955 Stage-1 map = 100%,  reduce = 0%, Cumulative CPU 2.32 sec
2026-04-23 00:44:48,291 Stage-1 map = 100%,  reduce = 100%, Cumulative CPU 3.95 sec
MapReduce Total cumulative CPU time: 3 seconds 950 msec
Ended Job = job_1776925353456_0002
MapReduce Jobs Launched:
Stage-Stage-1: Map: 1  Reduce: 1   Cumulative CPU: 3.95 sec   HDFS Read: 3407506 HDFS Write: 52 SUCCESS
Total MapReduce CPU Time Spent: 3 seconds 950 msec
OK
NULL    1
Movie   6131
TV Show 2676
William Wyler   1
type    1
Time taken: 22.701 seconds, Fetched: 5 row(s)
hive>
```

Step-06:- Top 5 Countries

- SELECT country, COUNT(*) AS total FROM netflix GROUP BY country ORDER BY total DESC LIMIT 5;

```
hive> SELECT country, COUNT(*) AS total
    > FROM netflix
    > GROUP BY country
    > ORDER BY total DESC
    > LIMIT 5;
Query ID = cloudera_20260423004949_7a8eebf9-5eed-4efa-9746-49a876495664
Total jobs = 2
Launching Job 1 out of 2
Number of reduce tasks not specified. Estimated from input data size: 1
In order to change the average load for a reducer (in bytes):
  set hive.exec.reducers.bytes.per.reducer=<number>
In order to limit the maximum number of reducers:
  set hive.exec.reducers.max=<number>
In order to set a constant number of reducers:
  set mapreduce.job.reduces=<number>
Starting Job = job_1776925353456_0003, Tracking URL = http://quickstart.cloudera:8088/proxy/application_1776925353456_0003/
Kill Command = /usr/lib/hadoop/bin/hadoop job  -kill job_1776925353456_0003
Hadoop job information for Stage-1: number of mappers: 1; number of reducers: 1
2026-04-23 00:49:30,756 Stage-1 map = 0%,  reduce = 0%
2026-04-23 00:49:37,046 Stage-1 map = 100%,  reduce = 0%, Cumulative CPU 2.15 sec
2026-04-23 00:49:44,464 Stage-1 map = 100%,  reduce = 100%, Cumulative CPU 5.89 sec
MapReduce Total cumulative CPU time: 5 seconds 890 msec
Ended Job = job_1776925353456_0003
Launching Job 2 out of 2
Number of reduce tasks determined at compile time: 1
In order to change the average load for a reducer (in bytes):
  set hive.exec.reducers.bytes.per.reducer=<number>
In order to limit the maximum number of reducers:
  set hive.exec.reducers.max=<number>
In order to set a constant number of reducers:
  set mapreduce.job.reduces=<number>
Starting Job = job_1776925353456_0004, Tracking URL = http://quickstart.cloudera:8088/proxy/application_1776925353456_0004/
Kill Command = /usr/lib/hadoop/bin/hadoop job  -kill job_1776925353456_0004
Hadoop job information for Stage-2: number of mappers: 1; number of reducers: 1
2026-04-23 00:49:51,633 Stage-2 map = 0%,  reduce = 0%
2026-04-23 00:49:56,957 Stage-2 map = 100%,  reduce = 0%, Cumulative CPU 1.83 sec
2026-04-23 00:50:02,150 Stage-2 map = 100%,  reduce = 100%, Cumulative CPU 3.43 sec
MapReduce Total cumulative CPU time: 3 seconds 430 msec
Ended Job = job_1776925353456_0004
MapReduce Jobs Launched:
Stage-Stage-1: Map: 1  Reduce: 1   Cumulative CPU: 5.89 sec   HDFS Read: 3406828 HDFS Write: 196056 SUCCESS
Stage-Stage-2: Map: 1  Reduce: 1   Cumulative CPU: 3.43 sec   HDFS Read: 200545 HDFS Write: 69 SUCCESS
Total MapReduce CPU Time Spent: 9 seconds 320 msec
OK
United States   669
        335
United Kingdom  172
India   49
"United States  44
Time taken: 38.636 seconds, Fetched: 5 row(s)
hive>
```

Step-07:- Content by Year

SELECT release_year, COUNT(*) FROM netflix
GROUP BY release_year ORDER BY release_year;

```
  set hive.exec.reducers.bytes.per.reducer=<number>
In order to limit the maximum number of reducers:
  set hive.exec.reducers.max=<number>
In order to set a constant number of reducers:
  set mapreduce.job.reduces=<number>
Starting Job = job_1776925353456_0005, Tracking URL = http://quickstart.cloudera:8088/proxy/application_1776925353456_0005/
Kill Command = /usr/lib/hadoop/bin/hadoop job  -kill job_1776925353456_0005
Hadoop job information for Stage-1: number of mappers: 1; number of reducers: 1
2026-04-23 00:58:10,504 Stage-1 map = 0%,  reduce = 0%
2026-04-23 00:58:22,985 Stage-1 map = 100%,  reduce = 0%, Cumulative CPU 1.75 sec
2026-04-23 00:58:29,471 Stage-1 map = 100%,  reduce = 100%, Cumulative CPU 3.29 sec
MapReduce Total cumulative CPU time: 3 seconds 290 msec
Ended Job = job_1776925353456_0005
Launching Job 2 out of 2
Number of reduce tasks determined at compile time: 1
In order to change the average load for a reducer (in bytes):
  set hive.exec.reducers.bytes.per.reducer=<number>
In order to limit the maximum number of reducers:
  set hive.exec.reducers.max=<number>
In order to set a constant number of reducers:
  set mapreduce.job.reduces=<number>
Starting Job = job_1776925353456_0006, Tracking URL = http://quickstart.cloudera:8088/proxy/application_1776925353456_0006/
Kill Command = /usr/lib/hadoop/bin/hadoop job  -kill job_1776925353456_0006
Hadoop job information for Stage-2: number of mappers: 1; number of reducers: 1
2026-04-23 00:58:39,720 Stage-2 map = 0%,  reduce = 0%
2026-04-23 00:58:52,790 Stage-2 map = 100%,  reduce = 0%, Cumulative CPU 3.42 sec
2026-04-23 00:59:19,742 Stage-2 map = 100%,  reduce = 100%, Cumulative CPU 10.69 sec
MapReduce Total cumulative CPU time: 10 seconds 690 msec
Ended Job = job_1776925353456_0006
MapReduce Jobs Launched:
Stage-Stage-1: Map: 1  Reduce: 1   Cumulative CPU: 3.29 sec   HDFS Read: 3406837 HDFS Write: 158 SUCCESS
Stage-Stage-2: Map: 1  Reduce: 1   Cumulative CPU: 10.69 sec   HDFS Read: 4512 HDFS Write: 22 SUCCESS
Total MapReduce CPU Time Spent: 13 seconds 980 msec
OK
NULL    8808
2010    1
2018    1
Time taken: 84.747 seconds, Fetched: 3 row(s)
hive>
```

- **Output:**

  - **Hive successfully displayed:**
    - **Movies vs Tv Shows**
    - **Top 5 Countries**
    - **Content by Year**

- **In this practical, Hive was used to analyze processed netflix_titles data and retrieve meaningful insights using SQL-like queries.**